

Figure: CampusRide v4.4 - Requirements Specification (R.1-R.12 anchored to HoQ EC1-EC12)

Status reflects the Railway-backed production snapshot of 2026-04-23 (184 users, 170 verified). Evidence column quotes deployed rows / files.

ID	EC	Requirement ("The system shall...")	Subsystem	Verification	Status	Evidence / Live trace
R.1	EC1	The system shall verify .edu institutional identity before granting access to any module.	Auth	Test	Deployed	170/184 verified (92.4%) at 2026-04-23
R.2	EC2	The system shall maintain bidirectional Socket.IO + REST channels secured by JWT.	Substrate	Test	Deployed	Express 5.1 + Socket.IO on Railway
R.3	EC3	The system shall display a verified-Cornell badge on every driver and rider profile card.	Carpool	Inspection	Deployed	ride/profile cards live in UI
R.4	EC4	The system shall enforce a 2-hour post-departure delay before opening the rating window.	Rating	Test	Deployed	RATING_READY_DELAY_MS const
R.5	EC5	The system shall persist bidirectional ratings keyed on (trip_id, rater_id, ratee_id).	Rating	Test	Deployed	schema enforced; 0 rows in prod
R.6	EC6	The system shall recompute users.avg_rating across ratings + activity_ratings on each rating UPSERT.	Rating	Test	Deployed	RPC recompute_user_rating
R.7	EC7	The system shall offer update-in-place rating revision (with documented dispute-window limitation).	Rating	Demo	Deployed	1 row per (trip,rater,ratee)
R.8	EC8	The system shall auto-create a ride-scoped group chat on first booking, expiring departure +1h.	Messaging	Test	Deployed	8 rating-reminder + 16 ride pushes
R.9	EC9	The system shall write to wxgroup_notice_record on createRide / marketplace_item / activity creation.	Outreach	Test	Deployed	82 rows: 62 mkt / 16 ride / 4 act
R.10	EC10	The system shall award/deduct points via point_transactions atomic with users.points RPC.	Points	Test	Specified	point_rules=0; table missing in prod
R.11	EC11	The system shall implement REPLY_REQUIRED guard on cold DMs (HTTP 403 until the recipient replies).	Messaging	Test	Deployed	context_type DM gating live
R.12	EC12	The system shall accept multi-institution domains via a configurable .edu whitelist.	Auth	Test	Pending	Cornell-only as of 2026-04